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REVIEW

The use of electrochemical impedance spectroscopy for biosensing

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Abstract This review introduces the basic concepts and terms associated with impedance and techniques of measuring impedance. The focus of this review is on the application of this transduction method for sensing purposes. Examples of its use in combination with enzymes, antibodies, DNA and with cells will be described. Important fields of application include immune and nucleic acid analysis. Special attention is devoted to the various electrode design and amplification schemes developed for sensitivity enhancement. Electrolyte insulator semiconductor (EIS) structures will be treated separately.

Keywords Impedance · Sensor · DNA · Antibodies · Cells · Enzymes

Introduction

Impedance spectroscopy is a powerful method of analysing the complex electrical resistance of a system and is sensitive to surface phenomena and changes of bulk properties. Thus, it is a valuable method in electrochemical research, where an entire conference series is devoted to progress in and applications of this technique (the EIS Symposium Series, the most recent of which took place in 2007 in Argeles sur Mer, France). It has been intensively used, for example, for the elucidation of corrosion mechanisms, the characterisation of charge transport across membranes and membrane/solution interfaces, and the optimisation of batteries. In the field of biosensors, it is particularly well-suited to the detection of binding events on the transducer surface. First examples of its use were reported at the end of the 1980s; however, the method has found increasing application in recent years due to advances made in instrumentation. Besides the detection of biorecognition processes, it is a valuable tool for characterising surface modifications, such as those that occur during the immobilisation of biomolecules on the transducer. In this review, a short introduction to the measuring principles of electrochemical impedance spectroscopy will be provided, followed by an overview of the usage of the technique in the area of biosensors. The technique has the inherent potential for label-free detection, which is of special interest in bioanalysis since this circumvents the need to modify biomolecules with fluorescence dyes, enzymes, redox or radioactive labels. Several examples of its application to the detection of DNA, antigens or antibodies have been demonstrated. However, an amplification step is often necessary to achieve a defined response with very low analyte concentrations. The different approaches used to obtain signal enhancement will be also treated within this review.

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Basic principles and terms in impedance spectroscopy

The impedance Z of a system is generally determined by applying a voltage perturbation with a small amplitude and detecting the current response. From this definition, the impedance Z is the quotient of the voltage–time function $V(t)$ and the resulting current–time function $I(t)$:

$$Z = \frac{V(t)}{I(t)} = \frac{V_0 \sin(2\Pi f t)}{I_0 \sin(2\Pi f t + \phi)} = \frac{1}{Y} \quad (1)$$

where V_0 and I_0 are the maximum voltage and current signals, f is the frequency, t the time, ϕ the phase shift

between the voltage–time and current–time functions, and Y is the complex conductance or admittance. The impedance is a complex value, since the current can differ not only in terms of the amplitude but it can also show a phase shift ϕ compared to the voltage–time function. Thus, the value can be described either by the modulus $|Z|$ and the phase shift ϕ or alternatively by the real part Z_R and the imaginary part Z_I of the impedance. This is illustrated in Fig. 1. Therefore the results of an impedance measurement can be illustrated in two different ways: using a Bode plot which plots $\log|Z|$ and ϕ as a function of $\log f$, or using a Nyquist plot which plots Z_R and Z_I.

The name impedance "spectroscopy" is derived from the fact that the impedance is generally determined at different frequencies rather than just one. Thus, an impedance spectrum is obtained that allows the characterisation of surfaces, layers or membranes as well as exchange and diffusion processes. To achieve this, the impedance spectrum is often analysed using an equivalent circuit. This circuit, which commonly consists of resistances and capacitances, represents the different physicochemical properties of the system under investigation [1–3]. Alternatively, the system can be described based on transfer functions derived from the basic laws of the processes involved, such as electrokinetics, diffusion, partition, etc.

However, it is not only possible to describe a system of interest, but the technique can also be used for analytical purpose. In this case, the change of one impedance element—a resistance or a capacitance—as a function of the solution composition is evaluated. In some cases it is also possible to correlate the overall impedance to a change in concentration. This can simplify measurements, since it is often sufficient to determine the impedance at just one selected frequency or within a limited frequency window (where the relative changes are largest) in such cases.

In electrochemical impedance spectroscopy, where the electrolyte solution is one component of the system to be

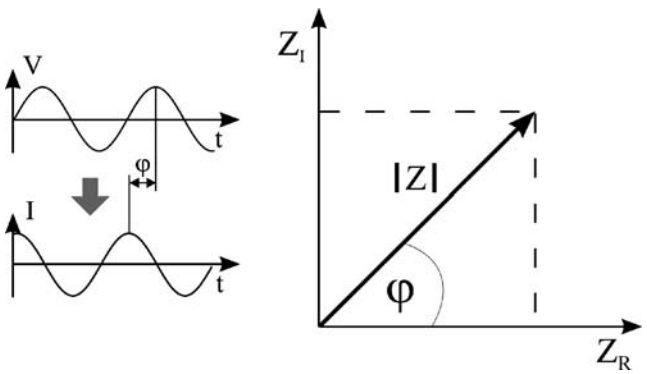


Fig. 1 Impedance is a complex value that is defined as the quotient of the voltage(time) and current(time) functions. It can be expressed as the modulus $|Z|$ and the phase angle ϕ, or it can be specified by the real (Z_R) and the imaginary (Z_I) parts of the impedance

Table 1 Definitions, frequency dependences and phase shifts of the impedance elements most often used to describe (bio)electrochemical systems

Impedance element	Definition	Phase angle	Frequency dependence
R	$Z=R$	0°	No
C	$Z_C = \frac{1}{j\omega C}$	90°	Yes
CPE	$Z_{CPE} = \frac{1}{A(j\omega)^{\alpha}}$	0–90°	Yes
W (infinite)[a]	$Z_W = \frac{\sigma}{\sqrt{\omega}}(1-j)$ $\sigma = \frac{RT}{n^2F^2\sqrt{2}}\left(\frac{1}{\sqrt{D_0 c_0}} + \frac{1}{\sqrt{D_R c_R}}\right)$	45°	Yes

[a] When the diffusion region is limited a different behaviour will be observed. A description of the two borderline cases with blocked and nonblocked transport at the end of the diffusion layer is given in [3])

investigated, four elements are usually used to describe the impedance behaviour: ohmic resistance, capacitance, constant-phase element and Warburg impedance. These elements and their definitions are summarised in Table 1. Equivalent circuits are used in order to approximate the experimental impedance data with these ideal or distributed impedance elements arranged in series and/or in parallel. Many electrochemical systems have been analysed according to this procedure. One can probably find a model for describing the impedance behavior of any particular system in the literature. This can be used at least as a starting point for analysis. For the situation of an electrode in contact with an electrolyte, the so-called Randles circuit is used (as shown in Fig. 2), comprising the solution resistance R_s, the charge transfer resistance R_{ct}, the double layer capacitance C_{dl} and the Warburg impedance W.

In the Nyquist plot shown in Fig. 2, the values for R_s and R_{ct} can be easily determined. The double layer capacitance can be calculated from the frequency at the maximum of the semicircle ($\omega=2\pi f=1/R_{ct}C_{dl}$). The product of R_{ct} and C_{dl} is often termed the time constant τ of the electrochemical process. The 45° line indicating Warburg-limited behaviour can be extrapolated to the real axis. The intercept is equal to $R_s + R_{ct} - 2\sigma C_{dl}$, from which σ and subsequently diffusion coefficients can be calculated (see Table 1). For analytical applications, however, the equivalent circuit is often simplified by neglecting the Warburg impedance. This can be done by choosing a frequency range where no 45° line is observed in the Nyquist plot and the interfacial or bulk impedance is dominant.

In order to characterise a biological material, e.g. antibodies or cells, electrodes must be introduced into the system, thus forming an electrochemical cell. Upon applying an ac voltage perturbation, the current is coerced to flow through all components of the system—the working electrode, the biological material, the solution and the counter electrode. The measured impedance is the sum of all of the individual contributions. Figure 3 provides a simplified equivalent circuit for this situation, neglecting the potential

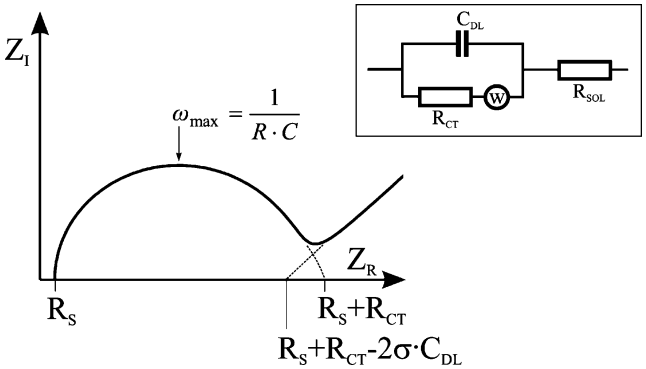


Fig. 2 Randles' equivalent circuit for an electrode in contact with an electrolyte. The double-layer capacitance C_{dl} results from charge being stored in the double layer at the interface. The charge transfer resistance R_{Ct} refers to current flow produced by redox reactions at the interface, and the Warburg impedance results from the impedance of the current due to diffusion from the bulk solution to the interface. R_s is the solution resistance afforded by the ion concentration and the cell geometry. R_s and R_{Ct} can easily be determined from the Nyquist plot, but also the σ value from the Warburg impedance can also be evaluated (see Table 1)

contribution of the Warburg impedance. For analytical applications two different situations can occur:

a) The impedance of the biological material is investigated as a function of the concentration of a certain analyte in solution or as a function of time. This is analogous to the analysis of the solution resistance as a function of ion concentration in solution (conductivity measurements). In both cases, the impedance of the two electrodes must be small compared to the impedance to be determined. This can be achieved by using electrodes with rather large surface areas. In addition, the nonspecific adsorption of biological material from solution must be avoided, since this would also increase the interfacial impedance. An example of this type of measurement is the determination of the cell mass during a fermentation process [4, 5]. The increasing density of cells within the solution results in an impedance change which can be easily monitored.

b) Alternatively, the biological component is immobilised on the working electrode and the interaction with an analyte molecule is detected. This is a typical biosensor application. Here the impedance of the sensing electrode (the working electrode modified with the biological component) controls the overall impedance. Thus, the impedance of the counter electrode must be significantly smaller. This is usually achieved by using an electrode with an area that is at least ten times larger than the sensing electrode. In addition, the surface can be roughened by electrochemical metal deposition or metal dissolution. By using a supporting electrolyte (0.01–1 M), the solution resistance can be adjusted such that it does not limit the current response.

When taking measurements with such surface-modified sensing electrodes, redox-active compounds are commonly added to the solution, resulting in a well-defined charge transfer resistance R_{ct}. If the redox-active compound is omitted or a blocking layer is applied to the electrode, rather capacitive impedance behaviour will be observed (since R_{ct} will become extremely large). Thus, a binding event at the electrode can be detected by following the change in R_{ct} in the first case or the change in the capacitance in the second case.

When performing electrochemical impedance spectroscopy, it is often necessary to control the situation at the working electrode. (i) In some cases the impedance can be measured at the "open circuit" potential (OCP). This means a situation which has equilibrated at the electrode in dependence on the solution composition. However, this situation must be stable during the measurements. Here, a simple two-electrode arrangement can often be used. (ii) Alternatively a dc potential must be fixed at the working electrode in order to define a situation at the electrode or to force a process to occur (e.g. an oxidation process). This is a typical potentiostatic mode, where a three-electrode arrangement is necessary. (iii) Another approach is to define a current to flow over the working electrode. This can be achieved in galvanostatic mode, again with a three-electrode arrangement.

Modern impedance analysers also contain data evaluation software and fitting programmes which allow the calculation of transfer functions according to an equivalent circuit. One can find models in the literature or one can choose an equivalent circuit from the implemented software as a starting point for the description of an individual system. In order to verify the validity of the chosen circuit, the quality of the fit to the experimental curve must be evaluated, and experimental parameters such as the elec-

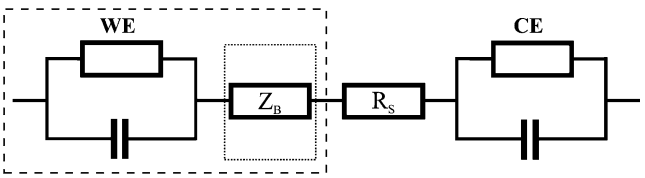


Fig. 3 Equivalent circuit for an electrochemical cell using two electrodes (working electrode, *WE*, and counter electrode, *CE*) and a biological component. Both electrodes are exemplified using a parallel circuit with a double layer capacitance and a charge transfer resistance. The solution resistance R_S and the impedance of the biological component Z_B are in series with the electrode impedance. The working electrode can be modified with the biocomponent and thus appears as the sensing electrode. This is indicated by the *dashed box*. For this type of measurement, the other impedance elements should be comparatively small. When the biological component needs to be investigated in solution, its value should determine the overall impedance to a large extent, and the impedances of the electrodes should be small. This type of investigation is indicated by the *dotted box*

trolyte or analyte concentrations, or a layer thickness or the electrode area must be varied.

The frequency range needed to determine the parameter of interest can also be deduced from such an analysis of the impedance data. For analytical applications, it is also possible to not only reduce the frequency range but also to limit the measurements to a few or even just one selected frequency. For example, this approach can be used when measuring chemically sensitive semiconductor structures that show highly capacitive behaviour [6, 7]. The frequency range that must be analysed varies with the system under investigation, but generally ranges from several tens of kilohertz to 10^{-1} or 10^{-3} Hz. Commercial instruments easily cover the range from 10^{-5} Hz to 10^{6} Hz. The impedance parameter of interest (e.g. the resistance or capacitance as a function of the concentration to be determined) can be obtained from a fit of the measured impedance data with a verified equivalent circuit, or the overall impedance can be directly correlated to the concentration.

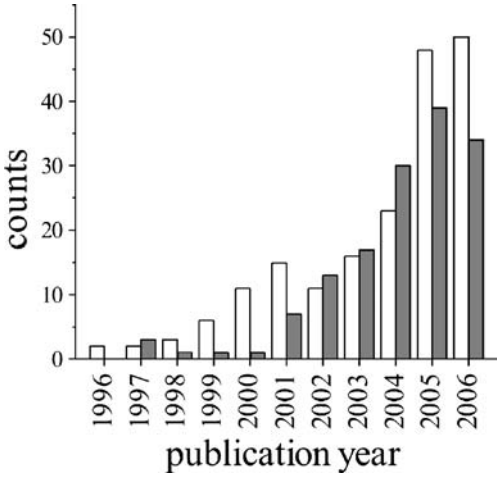


Fig. 4 Number of publications with the keywords "impedance spectroscopy" and "biosensor" over the last ten years. *Open* and *hatched bars* correspond to bibliographic searches of the ISI Web of Science and PubMed, respectively, performed using the keyword combination "impedance spectroscopy AND biosensor"

Impedimetric biosensors

Like all electrochemical biosensors, impedimetric sensors are bioelectronic devices that make use of the interactions of biomolecules with a conductive (or semiconducting) transducer surface. The detection process involves the formation of a recognition complex between the sensing biomolecule and the analyte at the interface of the electronic transducer, which directly or indirectly alters the electrical properties of the recognition surface. Because of the affordability and availability of impedance instrumentation nowadays, a trend towards the development of impedimetric biosensors appears to be underway. This can be seen from the increase in the number of publications on this topic over the past few years. Figure 4 shows the results of searches made in two different databases for the terms "impedance spectroscopy" and "biosensor", for publications since 1996.

Impedance spectroscopy is mainly applied in bioanalytics to monitor the biosensor fabrication process, but it is also used to investigate the recognition process at the biosensor. Applications have been demonstrated for various types of (bio)analytes, such as proteins, nucleic acids, whole cells, microorganisms, antibodies and antigens. Due to space limitations, a detailed and complete description of the different approaches cannot be given here; instead, the present review tries to provide an overview of the various strategies that have been used to design impedimetric biosensors, without claiming to be complete. The reader should note that alternative reviews on impedimetric biosensors are available [8–10].

The various efforts directed at specific and sensitive detection can be classified according to the electrode material (metals, metal oxides, glassy carbon, semiconductors, etc.), the electrode geometry (conventional electrode arrangement or interdigitated electrodes), the analyte (proteins, antibodies, nucleic acids, etc.), or according to the amplification protocol used (label-free, enzyme labels, conducting polymer films, nanoparticles, etc.).

Direct analyte detection

In order to construct a biosensor, the recognition element is immobilised on a single working electrode. Figure 5a shows the schematic structure of such an impedimetric biosensor. Different options exist for evaluating the recognition event at the sensor surface; besides the simple detection of the overall impedance at selected frequencies, one can analyse the changes in electrode capacitance or resistance.

The capacitance of the electrochemical double layer, C_{dl}, depends on all of the compounds present at the interface, which are predominantly solvent molecules, immobilised biomolecules, and optional films which promote the immobilisation and detection processes. A capacitance change is induced when the dielectric constant or the thickness of the double layer on the transducer surface changes. Both options have been investigated [11, 12]. When constructing a capacitive biosensor, the electrode surface is usually covered with an additional insulating layer to reduce Faradaic currents (i.e. to increase the charge transfer resistance in parallel to the capacitance). The biorecognition element is immobilised on top of this layer.

Capacitance measurements made in the absence of Faradaic currents also allow the influence of the electrical field on the biological recognition event to be investigated,

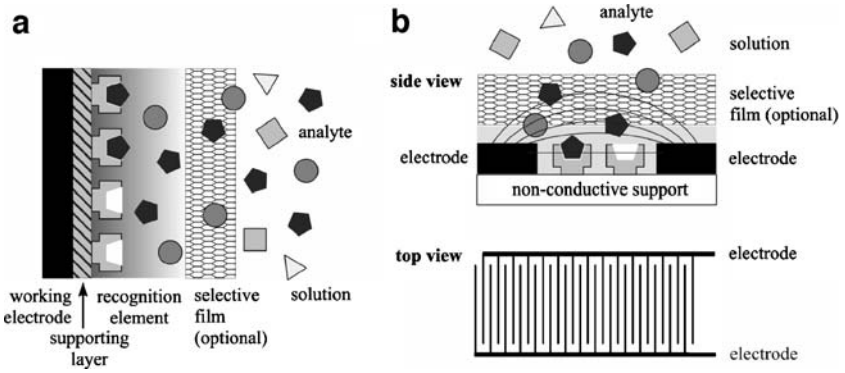


Fig. 5a–b Schemes of impedimetric biosensors using a modified working electrode as sensor (**a**) or an in-plane arrangement of two electrodes (interdigitated electrodes, *IDE*) (**b**). Immobilisation of the biocomponent occurs on the electrode in **a** and between and/or on the electrodes in **b**

by applying different dc working potentials [13]. The interaction between the recognition element and the analyte as well as their orientations can be influenced by the applied dc potential, particularly when the components of the interaction are charged.

An alternative approach to impedimetric detection of analyte binding is the determination of the charge transfer resistance, R_{ct}. These Faradaic impedance measurements are usually performed in the presence of a redox couple in solution and rely on changes in the barrier to redox conversion due to the formation of the recognition complex itself or a subsequent complex. The ferri-/ferrocyanide system is often chosen for this purpose. To achieve a defined and sensitive detection, impedance spectra are generally recorded for dc potentials equal to the open circuit potential (OCP) of the employed redox couple. The presence of a redox couple, however, is not absolutely necessary. Measurements performed in an inert electrolyte solution usually result in a much higher impedance, but as long as the impedance changes that take place during the biorecognition process are reproducible, this kind of measurement can also result in reliable data. This approach is particularly interesting when the conductivity of the recognition layer is changed by the specific binding event (this occurs, for example, when proteins are embedded in a lipid layer).

Self-assembled monolayers (SAM) are frequently used to promote the immobilisation of the biorecognition element. Such films are formed spontaneously by chemisorption and self-organisation from solution [14–17]. Long-chain thiols can effectively block the electrode surface from unwanted reactions and are thus often used in capacitive sensors. Since the SAM films do not have specific recognition properties, the biocompound is normally coupled to the surface with functionalised SAMs [18]. Alternatively, the SAM layer must be structured in order to create recognition sites. This can be termed as a kind of surface imprinting [19, 20]. An alternative method of obtaining well-ordered recognition layers is the transfer of a lipid bilayer (from liposomes) or a Langmuir–Blodgett (LB) film onto the electrode surface [21, 22]. Other immobilisation protocols rely on the preparation of thin polymer films, electropolymers or bioaffinity layers (e.g. avidin films capable of binding biotinylated biomolecules). More recently, polyelectrolyte films have been introduced for biomolecule immobilisation. This technique is based on the layer-by-layer adsorption of oppositely charged polymers, and provides an easy path to surface modification [23, 24].

In contrast to the use in signal transduction, impedance measurements have found wide acceptance in biosensor optimisation, and are mainly used to follow the different steps of surface modification during sensor preparation. This includes the formation of SAMs, Langmuir–Blodgett films, adsorption layers, as well as the binding of the recognition element to the transducer surface (see e.g. [25–28]).

Finally, it should be mentioned that there are also developments aimed at using impedance measurements for biochemical analysis without a recognition element. One such example is a study of its potential to detect glucose in a noninvasive way in body fluids [29].

Immune detection

As already implied, impedimetric analyte detection is frequently applied for immunosensing. One source of impetus for the development of sensors with impedimetric detection are the demands of point-of-care diagnostics, which require rather simple equipment and the potential for label-free analysis. Two immobilisation strategies are possible for immunosensing. The sensor chip can either be modified with antibodies that bind the respective antigen molecule, or the antigen itself can be immobilised, which then binds the complementary antibody. In both cases, the binding event results in a change in the electrical surface properties, although larger changes can be expected in the latter case because of the high molecular weight and the low dielectric constant of the antibodies.

Initially capacitive sensors were preferably used for the detection of immune reactions [14, 30–32]. Charge transfer

can be diminished by using long-chain SAM layers for example. When used in combination with a potential-jump method, such sensors can yield very sensitive measurements [31, 33]. It has been also shown that a continuous binding analysis is possible, allowing a mathematical description of the binding behaviour to be obtained [34].

In order to reduce the influence of nonspecific binding, which influences the capacitance in a similar way to the specific interaction, a differential mode of measurement was proposed [35]. Antibodies can be incorporated into Langmuir–Blodgett films while retaining their binding properties. This can result in very ordered recognition structures [21, 36]. Very sensitive capacitive measurements have been demonstrated in a flow system for interleukin-6 using an epoxysilane functionalisation [37] or for interferon γ with a SAM modification for antibody immobilisation [38].

Resistance-based sensors have been also developed [39–41]. For example, the human mammary tumor associated glycoprotein was detected by specific antibodies immobilised on gold by spontaneous adsorption. The binding of the complementary antigen resulted in a change in the charge transfer resistance, R_{ct} [40]. The resistance-based detection was also used to study receptor–ligand interactions and allowed the detection of, for example, odorant molecules [42]. Another system for immune analysis used ultrathin platinum layers and evaluated the conductivity changes based on an impedance model analysis [41]. However, often the sensitivity of direct binding detection is insufficient to meet practical demands. Thus, several amplification strategies have been developed, as will be discussed in the next section.

An enhancement in sensitivity can be already obtained by applying a special approach for immobilising the recognition element using conducting polymers such as polypyrrol. This can help to enhance the response, since the conductivity of the polymeric network is strongly influenced by conformational changes induced by binding events [43, 44]. Biotinylated polypyrrol films were used to immobilise biotinylated antibodies via avidin [45, 46]. Antigen binding was linked to an increase in the charge transfer resistance. The respective detection limit was 10 pg mL^{-1} of human IgG.

In a similar direction, systems have been reported where the change in ion conductivity of a lipid bilayer with incorporated ion channels is applied for antibody–antigen binding detection [47, 48].

Nucleic acid detection

Besides immunosensing, the impedimetric detection of nucleic acids is being researched. ssDNA and dsDNA have been discriminated using a hanging mercury drop electrode [49, 50]. Desorption of denatured ssDNA was accompanied by a larger dielectric loss than occurred for the desorption of native dsDNA. More interesting for sensing applications are investigations on gold electrodes. Several sensor systems have been reported for a voltammetric analysis of hybridisation [51–55]. However, here the addition or the coupling of a redox indicator is required. The presence of ss- or ds-DNA on the electrode surface was found to result in different molecular structures, which can be evaluated by different spectroscopic and microscopic methods [56]. This provides the basis for the impedimetric detection of oligonucleotide concentrations and single base pair mismatches. For DNA detection in particular, the charge accumulation during hybridisation was used to alter the conversion of a redox-active substance [57, 58]. Intercalation can be used to modify the sensitivity [59]. The surface concentration of the probe DNA has a significant influence on the sensor performance. It can be drastically increased by using dendrimers [60]. Screen-printed electrodes modified with a polyelectrolyte are an alternative and have been used with ss-PCR products as capture molecules [61]. Instead of capture-DNA, PNA can be used, resulting in a much higher binding force and stronger charge differences after target DNA binding [62]. Capacitive measurements were found to be less sensitive than resistance-based methods [63].

One particular task in DNA analysis is the detection of single base pair mismatches. Impedance spectroscopy can be used for this task too [64, 58]. The transduction method can be nicely combined with specific biochemical protocols such as ligation [65]. In general, it is possible to analyse different DNA structures, such as B- or M-DNA conformations on a sensor surface, impedimetrically [66].

Analogously to immune detection using conducting polymers, probe DNA can be linked to such polymer films to evaluate the change in interface properties after target DNA binding. Here an increase in resistance was observed [67–69]. Streptavidin can be copolymerised with pyrrol, resulting in a simple immobilisation scheme for the biotinylated probe DNA [70]. Functionalised polypyrrol derivatives with unsaturated side chains were shown to be advantageous in sensor applications [71]. The technique of electropolymerisation is particularly well-suited to the immobilisation of DNA upon microelectrodes and thus array fabrication [72]. As an alternative to polypyrrole, a quinone-containing polymer was developed [73] or polythiophene was used [74]. The use of changes in conductivity upon hybridisation has also been demonstrated using nanoparticle layers of praseodymium oxide and covalently fixed ss-DNA [75].

As well as the detection of DNA–DNA binding, the impedance technique can also be used for the analysis of DNA–analyte interactions. Several small organic molecules and large proteins show a specific interaction with DNA

duplexes. This can be used for analytical determinations of these substances. Examples include cisplatin [76] and mythramycin [77], as well as specific DNA–binding proteins [78]. Aptamers have appeared as a new promising class of recognition molecules, have appeared, and have also been used in combination with impedimetric transduction [78–82]. The resistance change upon analyte binding is often evaluated [80–83].

MIPs, enzymes and receptors

Molecular imprinted polymers provide a surface which acts as a "negative" of the analyte and can thus recognize the analyte within a mixture. Because of their chemical nature they can be reproducibly fabricated in different formats and are stable for long periods in different environments. Besides their application to chromatographic separations, interesting examples of their applications in sensors have been reported. Thin surface films are particularly suited to impedimetric transduction of the binding event. Thus, capacitive sensors have been developed for organic molecules [19, 84] as well as for whole cells and viruses [85, 86].

In the area of enzyme sensors, amperometric electrodes are by far the most attractive transducers. However, the change in the redox conversion of a redox-active compound due to the turnover of an enzyme (in the presence of the respective substrate) can also be detected by following the charge transfer resistance. This has been demonstrated for glucose detection using glucose oxidase and parabenzoquinone as mediator [87].

As well as determining substrates and inhibitors of enzymes, the enzyme activity itself has also often been analysed. An interesting approach that used degradable polymer films was introduced. These polymers can be coated onto an electrode, and degradation through a biocatalytic reaction can occur directly due to the enzyme acting on the polymer chains, or indirectly due to products of the enzymatic conversion. The diminished film thickness is easily followed by impedance measurements. This has been shown for the detection of urea [88, 89], glucose [90] and enzymes such as chymotrypsin or lipase [91]. This scheme can be nicely combined with immunoassays using enzymes as labels [88].

Another way in which impedance is being applied for sensing involves incorporating a "signalling" protein in a lipid layer on an electrode and then evaluating the conformational changes upon its interaction with a specific analyte. Often this is accompanied by changes in ion mobility through the surface-confined film. Thus, conductivity changes can be analysed. This principle has been nicely demonstrated with gramicidin-coupled antibody fragments within a lipid membrane [92]. One example of the use of sensing substrates and inhibitors is given by Naumann et al. [93]. The authors used peptide-tethered lipid membranes on a gold support with cytochrome c oxidase and studied proton transport across the membrane. Impedance spectra could be correlated to the concentrations of inhibitor and substrate of the enzyme. Other examples are the detection of paraoxone [94], quinacrine [95], or the bacterial toxin streptolysin [96].

In general, conformational changes of recognition molecules upon binding are very applicable to analytical detection by impedance measurements. An example is glucose detection using the galactose/glucose receptor protein. The change in the effective protein layer thickness upon binding was followed by impedance analysis using an appropriate electrode design in the form of nanopores [97].

Cells

Further recent developments are cell-based assays for the detection of specific analytes or for the screening of large libraries of substances for their biological effects. This relies on the concept of effect monitoring. Here, parameters are needed that report the response of the whole biological system with respect to the stimulus under investigation. Impedance is one of these parameters because it can be used for the indirect detection of metabolic activity, cell adhesion on surfaces, response to potential drugs and for cytotoxicity tests [98, 99]. Studies have been reported in which the status of cell cultures [100–103] or even single cells [104] has been analysed. For example, apoptosis-induced changes in human colon cancer HT-29 cell shape have been investigated [105].

Impedance is not only suitable for obtaining information on the status of surface-fixed cells; it can also be used to detect the number of cells in solution. This has been intensively investigated in microbiology and has been used for the detection, quantification, and identification of bacteria [102, 106–109], as well as in micromachined devices for cell counting [4, 110, 111] or cell differentiation [5]. For the specific capture of bacterial cells, antibodies or bacteriophages can be used [112]. Impedimetric analysis has also been applied to the detection of leukaemia cells [113, 114].

Amplification of impedimetric signals

Despite the potential for direct binding detection, many interfacial impedimetric assays suffer from the disadvantage that the generated signals (changes in C_{dl} and/or R_{ct}) are relatively small. This has led to the development of several amplification strategies which can be grouped according to the part of the sensing system that they rely on: immobilisation, biochemical recognition or transducing electrode.

Immobilisation and biochemical strategies

In order to increase the change in capacitance due to an altered dielectric constant, polymeric membranes were used for recognition element fixation. Indeed, the incorporation of antibodies into a membrane induced a change of C_{dl}. As an example, Maupas et al. [35] have modified platinum electrodes with different polymeric layers and used them in a flow injection system for real-time immune detection. As mentioned above, a conducting polymer layer can also be usefully applied for enhanced signal generation [43, 67, 69].

A smart amplification approach was taken from classical immuno-assays, which includes the application of enzyme labels bound to a component of the sensing system. This results in a sandwich-like structure. For example, Bresler et al. have used a capacitive sensor structure (based on passivated interdigitated electrodes) and combined the antigen–antibody reaction with a second binding event, thus coupling catalase to the surface. Catalase converts H_2O_2 to oxygen, and the gas bubbles formed near the surface change the dielectric constant drastically, thus allowing very sensitive detection, as shown for HIV antibodies for example [115].

The enzyme label can also catalyse the formation of an insoluble product which precipitates onto the electrode surface and amplifies the changes in R_{ct} and C_{dl} [116–121]. This principle was introduced for the detection of DNA [116] and antibodies against dinitrophenyl at an electrode surface functionalised with the SAM technique [117]. Horseradish peroxidase (HRP) was used as enzyme label. The technique can be also applied for the detection of autoantibodies in physiological samples [122] or antibodies against the hepatitis B surface antigen (HBsAg) [123]. Screen-printed electrodes have been found to be suitable transducers for this kind of amplification [27].

The HRP label was also used for DNA detection of the Tay–Sachs genetic disorder with high sensitivity [124]. Recently the potential of this method has been demonstrated in comparison with a label-free DNA analysis [125]. Further applications of this amplification method, but with alkaline phosphatase used as label, have been demonstrated for the analysis of viral RNA and single nucleotide polymorphisms [126, 127]. Amplified DNA detection was achieved using liposomes labelled with multiple HRP tags, which resulted in an amplification factor of approximately 10^5 [120].

An enzyme label was also used for the impedimetric detection of bacteria cells of *Escherichia coli* [128]. The electrode surface was initially modified with an *E.coli* antibody. The binding of the cells were subsequently detected with faradaic impedance measurements. Signal amplification was achieved using a secondary antibody against *E.coli*, which was modified with alkaline phosphatase, catalysing the oxidation of 5-bromo-4-chloro-3-indolyl phosphate and thus yielding the insoluble indigo derivate. The detection limit was 6000 cells mL^{-1}. This amplification technique enables the detection of a very low surface coverage of antibody–antigen complexes. However, regeneration of the sensor surface is difficult if not almost impossible, and impedes the repeated usage of the sensor.

A way around this problem might be the use of nanoparticles, which have also been applied for signal generation and amplification [129–131]. Two different approaches can be discerned. First, nanoparticles have been used for electrode modification before binding of the recognition element. In this case it has been shown that the enhanced surface concentration is mainly responsible for the signal increase [132]. In the second approach, the particles have been applied as labels of biomolecules. Examples of the first approach are gold nanoparticles [77, 133–135] or multiwall carbon nanotubes [136], and examples of the second one include quantum dots [129] coupled to oligonucleotides or gold-labelled antibodies or DNA [83, 137, 138]. In this case, the interfacial properties are also altered and thus the conversion of the negatively charged redox compound ferri-/ferrocyanide is changed at the electrode surface. The nanoparticle approach can also be combined with surface modification by polypyrrole [136, 139].

A very simple amplification route has been introduced by the Willner group [140, 141]. Binding of negatively charged liposomes to the recognition surface can drastically alter the charge transfer of a redox couple, particularly when it possesses the same charge as, say, ferri-/ferrocyanide. In this scheme, the DNA hybridisation is followed by one or several secondary binding steps, resulting in an accumulation of highly charged liposomes on the electrode. Alternatively, the primary binding event can be amplified by the formation of a biotin–streptavidin network on the surface. This was shown for antibody–antigen interaction analysis [142].

Electrode design

As well as biochemical amplification, the transducer itself also provides the potential for further sensitivity enhancement. Instead of using a single working electrode, an in-plane geometry can be applied, which makes use of two closely spaced electrodes. These electrodes are termed “interdigitated electrodes” (IDEs) and can be formed from different metals [143–146]. The geometry takes advantage of the changed conditions observed for the current flow occurring mainly very close to the surface and which therefore shows a much higher sensitivity to surface changes compared to the conventional design. The biorecognition element can be immobilised on both electrodes, in the gap between the electrodes or on top of both. This electrode

design has an additional advantage: by preparing the electrodes on a chip, the whole electrochemical setup is already integrated and no external electrodes are needed. The basic scheme is shown in Fig. 5b. A highly sensitive sensor response requires strong electrical fields, which can be achieved by keeping the electrodes close together [147]. Typical distances between the electrode fingers are in the range of 1–10 μm and are limited by the fabrication procedure. However, electrode widths ranging from 500 nm [148] to less than 200 nm have been achieved so far. A theoretical analysis demonstrates that 80% of the current between electrodes separated by about 250 nm will flow in a layer no higher than 250 nm above the surface. This may indicate that the reproducible production of interdigitated electrodes with dimensions in the nm range will significantly increase the sensitivity for direct binding analysis.

Immune detection using IDEs takes advantage of the change in resistance close to the surface due to protein accumulation and the change in the dielectric properties due to the presence of substances with lower dielectric constants than water. Thus, label-free detection becomes feasible [148–150]. IDEs can be prepared on polymer substrates, offering the potential for disposable sensor production [151].

IDE-based impedance measurements can be also applied to the detection of DNA hybridisation without the need for a label. This relies mainly on the change in charge accumulation after double strand formation [145, 146, 148, 152, 153]. This is also valid for the detection of aptamer binding [154]. Furthermore, signal enhancement strategies such as polymer films for immobilisation or the application of nanoparticles can be combined with IDEs for sensitive detection [155, 156].

Receptor–ligand binding is often accompanied by structural changes in the complex, as mentioned in the previous section. This can be used for sensitive ligand detection, particularly when a receptor which is a chemically gated transport protein is used, such as the nicotinic acetylcholine receptor. When incorporated in a lipid membrane, acteylcholine and receptor antagonists can be measured by capacitance changes [157].

IDE-based sensors have also found application in enzyme detection. For example, an enzyme-based impedimetric sensor was developed for the detection of collagenase. Impedance changes were induced by the proteolytic digestion of gelatine-coated interdigitated electrodes [158]. Degradation of the layer produced a rapid increase in the impedance. Other examples based on the enzyme-catalysed polymer transformation include sensors for urea [159–161] or creatinine [160].

The interdigitated electrode design was also used for the analysis of cells in suspension [162–164] and for the detection of cells specifically captured by immune reactions [165, 166].

As shown for single working electrodes, IDE-based systems can also be combined with different biochemical amplification schemes. For example, DNA hybridisation can be extremely sensitively detected by using Au nanoparticles coupled to an oligonucleotide. The catalytic properties of the nanoparticle surface can be used for silver deposition and thus to enlarge the particles. When this enlargement proceeds to an extent that the surface-bound particles contact each other then a conductive pathway is formed between two electrodes [167, 168], resulting in an ultrasensitive detection scheme. An alternative approach to sensitivity enhancement in immunosensing uses conducting polymer chains coupled to the antigen [169]. After the binding of the modified antigen to the recognition surface, the conductance of the gap increases and sensitive detection becomes possible. The same effect can be induced if an enzyme label is coupled to the antigen. In this case, the sensor surface can be additionally covered with a membrane that is permeable to iodine, which is produced by the enzyme label HRP. After binding the antigen HRP conjugate, iodine is generated from iodide and hydrogen peroxide, which leads to increased conductance of the promoting film, consisting of tetra-tert-butyl copper phthalocyanine within the gap [170].

As an alternative or a complement to IDEs, pore-like electrodes have been developed. These electrodes rely mainly on the very sensitive detection of conductivity changes within the pores by biochemical reactions. Examples have been demonstrated for bacterial binding proteins [171] as well as for cell assays [172].

Semiconductor devices

A special field in impedance-related sensors focuses on semiconductor-based structures. These are mainly semiconductor/insulator structures that detect chemical changes at the insulator surface that is in contact with the solution by analysing the semiconductor/insulator interface. The classical ion-selective field effect transistor (ISFET) can be combined with enzymes or antibodies, yielding enzyme FETs (ENFETs) or immuno-FETs (IMFETs), which will not be discussed in this review [173, 191, 192]. However, another FET-derived, but simpler, structure—the electrolyte–insulator–semiconductor or EIS structure—is characterised by capacitance measurements and will therefore be discussed here. For a long time EIS structures were only seen as an intermediate step towards ISFET development, but they have recently gained attention as a separate sensor platform, mainly due to their ease of preparation and progress in instrumentation for detecting capacitance changes.

Enzymes can be effectively coupled to ion-sensitive semiconductor structures when the enzyme reaction changes the ion concentration near the surface. When the semiconductor structure carries the respective ion-sensitive

layer then an interfacial potential is generated, affecting the space charge region in the semiconductor, which can be analysed by following the capacitance of the structure. For EIS structures, examples are the detection of penicillin using penicillinase [174], formaldehyde with formaldehyde dehydrogenase [175] or organophosphorus pesticides using the respective hydrolase [176]. An increase in surface area has been shown to be advantageous to analytical performance [177]. Very often the shift in the capacitance–voltage curve along the voltage axis is evaluated rather than the change in impedance due to the biocatalysis. Also in this area, impedance measurements are used to characterise the immobilisation steps (e.g. [178, 179]).

There have also been several attempts to combine immune reactions with semiconductor-based sensors. However, the formation of the immune complex often does not enhance the overall impedance or change the charge situation on the gate. Moreover, for the detection of charged proteins, the measurements are additionally sensitive to the ionic strength of the solution. Using immobilised antibodies on Si/SiO_2 surfaces, the antigen α-fetoprotein was detected down to 1 ng mL^{-1} (15 pM) [180]. A similar approach was used to detect enterotoxin. A capacitance change was observed after incubation with 10 μg mL^{-1} (0.4 μM) [181].

Because of the high charge in a DNA strand, semiconductive heterostructures have been succesfully used to detect nucleic acids. The charge accumulation during hybridisation results in a change in the flat band potential which can be easily detected by capacitance measurements [182–186]. However, different sensitivities have been reported, and not all of the factors that determine the signal output are fully understood. Thus, recent investigations have concentrated on elucidating these parameters and modelling the response of the device [187, 188].

In the field of semiconductor devices, many efforts aimed at overcoming practical limitations, such as the limited sensitivity, are underway. One direction can be seen in the use of modified device structures and the application of nanowires [189, 190]. Although high sensitivities have been reported, reproducible production and operational stability are still targets that are yet to be attained.

Conclusions

The various examples in the literature illustrate the potential of the impedance technique for analytical purposes. It can be stated that impedimetric detection can achieve the sensitivity required for practical applications in medicine, gene analysis, food industry or environmental analysis. Interdigitated electrodes and amplification protocols contribute enormously to the present state-of-the-art. However, many of the sensors described in the literature have not been tested with real samples. Additional problems are expected when interfering compounds are present and nonspecific binding occurs. This is much more severe for impedance analysis than for optical or amperometric detection, and will increase the reported detection limits. Furthermore, the recording of a full impedance spectrum covering a broad range of frequencies is time-consuming but not always needed. Thus, until now impedimetric techniques have been preferentially used as a method of characterisation rather than as an analytical method for daily routine applications. However, label-free and direct binding detection is such an attractive feature that the door to analytical applications has opened wide. Therefore, it can be expected that further research will be directed at simplifying measurement schemes and devices (e.g. single or dual frequency measurements), robust amplification cycles and applications performed under real conditions.

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